Course

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1.1.1. Calculate Momentum

05:06

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).

Input Format:
- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:
- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases +

calculate... Submit

```
mass = float(input())
velocity = float(input())
momentum = mass * velocity
print(f"{momentum:.2f}kgm/s")
```

Average time 0.004 s 3.50 ms

Maximum time 0.005 s 5.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 5 ms Debug

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

Test case 2 3 ms

Terminal Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

28:55

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:
$1 \le n \le 999$

Input Format:
- The input consists of a single integer n.

Output Format:
- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases +

condition... Submit

```python
import math
number = int(input())
if 1 <= number <= 9: #Single-digit number
    print(number ** 2)
elif 10 <= number<=99: # Twodigit number 6
    print(f"{math.sqrt(number):.2f}")
elif 100<= number<=999: # Threedigit number
    print(f"{math.cbrt(number):.2f}")
else:
    print("Invalid")
```

Average time 0.002 s 1.71 ms
Maximum time 0.003 s 3.00 ms
4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 3 ms Debug

Expected output	Actual output
9	9
81	81

Test case 2 2 ms
Test case 3 2 ms

Terminal Test cases

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:
- The first line contains the first date in the format $YYYY - MM - DD$.
- The second line contains the second date in the format $YYYY - MM - DD$.

Output Format:
- An integer representing the number of days between the given dates.

Note:
- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```python
from datetime import date
from datetime import datetime

# Input dates
date1 = input()   # first date
date2 = input()   # second date

# Convert string to date format
d1 = datetime.strptime(date1, "%Y-%m-%d")
d2 = datetime.strptime(date2, "%Y-%m-%d")

# Calculate difference
difference = d2 - d1

# Print number of days
print(difference.days)
```

Average time 0.006 s 6.25 ms | Maximum time 0.019 s 19.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 19 ms Debug

Expected output	Actual output
2014-07-02	2014-07-02
2014-11-02	2014-11-02
123	123

Test case 2 2 ms

Terminal Test cases

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```python
#write your code here...
num=int(input())
reversed_num=int(str(num)[::-1])
print(reversed_num)
```

Average time	Maximum time
0.002 s	0.003 s
1.60 ms	3.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 3 ms Debug

Expected output	Actual output
5367	5367
7635	7635

Test case 2 2 ms

Terminal Test cases

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1.1.5. Multiplication Table

00:17

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases +

multiplica... Submit

```python
#write your code here...
num=int(input())
for i in range(1,11):
    print(f"{num} x {i} = {num*i}")
```

Average time	Maximum time
0.002 s	0.004 s
1.75 ms	4.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40

Terminal Test cases

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1.2.1. Pass or Fail

00:38

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases +

passorFa... Submit

```python
n = int(input())

marks = list(map(int, input().split()))
fail = False
for m in marks:
    if m<40:
        fail = True
        break
if fail:
    print("Fail")

else:
    percentage = sum(marks) / n
    print("Aggregate Percentage: {:.2f}".format(percentage))

    if percentage > 75:
        print("Grade: Distinction")
    elif percentage >=60:
        print("Grade: First Division")
```

Average time 0.003 s 2.60 ms

Maximum time 0.006 s 6.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 6 ms Debug

Expected output	Actual output
4	4
55 65 78 89	55 65 78 89
Aggregate Percentage: 71.75	Aggregate Percentage: 71.75
Grade: First Division	Grade: First Division

Test case 2

Terminal Test cases

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1.2.2. Fibonacci Series

01:54

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases +

fibonacci....

Explorer Submit Debugger

```python
def fibonacci(n):
    if n == 1:
        return 0
    if n==2:
        return 1
    else:
        return fibonacci(n-1)+fibonacci(n-2)


n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")

```

Average time	Maximum time
0.006 s	0.018 s
6.25 ms	18.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 2 ms Debug

Expected output	Actual output
5	5
0 1 1 2 3	0 1 1 2 3

Test case 2 2 ms

Terminal Test cases

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1.2.3. Pattern - 1

00:10

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

rightangl... Submit

```python
n = int(input())


for i in range(1, n + 1):
    print("* " * i)
```

Average time	Maximum time
0.002 s	0.002 s
1.83 ms	2.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 2 ms Debug

Expected output	Actual output
5	5
*	*
* *	* *
* * *	* * *
* * * *	* * * *
* * * * *	* * * * *

Terminal Test cases

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1.2.4. Pattern - 2

00:07

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

numberP... Submit

```python
n = int(input())

for i in range(1, n + 1):
    
    for j in range(1, i + 1):
        print(j, end=" ")
    print()

```

Average time	Maximum time
0.003 s	0.004 s
2.75 ms	4.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output	Actual output
5	5
1	1
1 2	1 2
1 2 3	1 2 3
1 2 3 4	1 2 3 4
1 2 3 4 5	1 2 3 4 5

Terminal Test cases

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